

A customer support chatbot performs well on short conversations but starts generating inconsistent responses during lengthy interactions. Investigation reveals that the model struggles to maintain context across long chat histories. How would you address this challenge using your understanding of Transformer architecture?

- Long conversations can exceed the model's effective context window, causing loss of important information.
- Analyze attention patterns to determine whether earlier messages receive meaningful attention.
- Implement conversation summarization for older dialogue turns.
- Maintain structured memory containing key entities, preferences, and decisions.
- Use retrieval-based context selection to provide only relevant history.
- Consider long-context Transformer architectures if supported.
- Monitor response consistency and context retention metrics.
- The objective is to preserve critical conversation history while controlling computational costs.

